

Showcase: Parsing Classical Chinese Texts

The Jupyter Notebook is a showcase on how to use Yasuoka BERT models to parse Classical Chinese texts.

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```
import pandas as pd
from transformers import pipeline
```

Use Yasuoka BERT models to parse Classical Chinese texts

First, load the Yasuoka BERT models.

```
nlp=pipeline("universal-dependencies",

             # will download the model at the first use
             "KoichiYasuoka/roberta-classical-chinese-large-ud-goeswith",

             # Or download it manually, and use it from a local path
             # "./roberta-classical-chinese-large-ud-goeswith",

             trust_remote_code=True,
             aggregation_strategy="simple")
# device = 0) # 0: cpu, 1: gpu
```

Device set to use mps:0

Then, define a function to parse a string.

The function will return a dataframe containing columns as follows:

['token_id', 'token', 'token_pos', 'token_feats', 'head_id', 'dependency_relation']

```
# define a function to parse a string
# return a dataframe

def parse_text(mytxt:str):

    myresults = nlp(mytxt)

    myresults2 = myresults.split('\n')
```

```

# delete the first line
del myresults2[0]

# delete the last two lines
del myresults2[-1]
del myresults2[-1]

mydf = pd.DataFrame([line.split("\t") for line in myresults2]).iloc[:,
[0,2,3,5,6,7]]
mydf.columns = ['token_id', 'token', 'token_pos', 'token_feats', 'head_id',
'dependency_relation']

return mydf

```

See the following example.

```

# Note that when you a Yasuoka model to parse a Classical Chinese string,
# you may need to first split the string into sentences and
# remove all punctuation marks.
# Then parse each sentence separately.
# The following is an example of the parsing of a single sentence.

my_sentence = "太祖姓楊名行密字化源廬州合肥人也"

parse_text(my_sentence)

```

	token_id	token	token_pos	token_feats	head_id	dependency_relation
0	1	太	VERB	Degree=Pos VerbForm=Part	2	amod
1	2	祖	NOUN	—	3	nmod
2	3	姓	NOUN	—	8	nsubj:outer
3	4	楊	PROPN	NameType=Sur	12	nsubj
4	5	名	NOUN	—	4	flat
5	6	行密	PROPN	NameType=Giv	4	flat
6	7	字	NOUN	—	8	nsubj
7	8	化源	PROPN	NameType=Giv	12	nsubj
8	9	廬	PROPN	Case=Loc Name-Type=Geo	10	nmod
9	10	州	NOUN	Case=Loc	11	nmod

	token_id	token	token_pos	token_feats	head_id	dependency_relation
10	11	合肥	PROP	Case=Loc Name- Type=Geo	12	nmod
11	12	人	NOUN	—	0	root
12	13	也	PART	—	12	discourse:sp